

A disproof of Conjecture 00000001213

In Sylver coinage the prime 5 is a winning first move

1 The conjecture

Conjecture 00000001213 reads:

Definition: Sylver Coinage: two players alternately name positive integers that may not be nonnegative integer combinations of previously named ones; naming 1 loses.

Conjecture: The first player wins, and every winning first move is composite; no prime first move is winning — after a prime opening the second player has a winning response.

The conjecture makes three claims:

- (C1) the first player wins;
- (C2) every winning first move is composite;
- (C3) no prime first move is winning, i.e. after a prime opening the second player has a winning response.

We prove that (C1) is true and that (C2) and (C3) are false. Indeed, every prime $p \geq 5$ is a winning first move. This is Hutchings' theorem; we give a complete proof and formalize it in Lean 4.

2 Definitions

A *position* is the finite set S of numbers named so far. A number n is *legal* after S if $n > 0$ and $n \notin \langle S \rangle$, where $\langle S \rangle$ is the additive submonoid of $\mathbb{N}$ generated by S , i.e. the set of nonnegative integer combinations of S . Game values are defined inductively, as in the accepted solution of Conjecture 00000008869:

- the player to move from S *wins* if some legal $n \neq 1$ leads to a position $S \cup \{n\}$ that the next player loses;
- the player to move from S *loses* if every legal $n \neq 1$ leads to a position that the next player wins.

Naming 1 loses at once, so it is never a winning move. If 1 is the only legal move, the player to move loses. A first move n is *winning* if n is legal after $\emptyset$, $n \neq 1$, and the second player loses from $\{n\}$. A number is *composite* if it is greater than 1 and not prime.

3 The proof

Lemma 1 (Determinacy). *If every legal move after S is smaller than B , then the player to move from S either wins or loses.*

Proof. Use induction on the number of legal moves below B . Naming a legal n removes n from the legal set and adds nothing to it. If some legal $n \neq 1$ leads to a lost position, the mover wins.

Otherwise every successor position is determined by induction and is not lost, so it is won, and the mover loses. $\square$

Lemma 2 (Closure invariance). *If $\langle S \rangle = \langle T \rangle$, then S and T have the same game value.*

Proof. Legality depends only on $\langle S \rangle$, and $\langle S \cup \{n\} \rangle = \langle n \rangle + \langle S \rangle$. Now use induction on the game value. $\square$

Lemma 3 (Symmetry). *Let $\gcd(p, q) = 1$ with $p, q > 1$, and let $t = pq - p - q$. If $x \notin \langle p, q \rangle$, then $x \leq t$ and $t - x \in \langle p, q \rangle$.*

Proof. The number t is the Frobenius number of $\langle p, q \rangle$: the largest integer not in $\langle p, q \rangle$. This is Mathlib’s `frobeniusNumber_pair`. Hence $x \leq t$, and $x + pq > t$ lies in $\langle p, q \rangle$. Write $x + pq = aq + bp$ with $a, b \geq 0$. If $a \geq p$, then $x = (a - p)q + bp \in \langle p, q \rangle$, which is impossible. So $a \leq p - 1$, and likewise $b \leq q - 1$. Then

$$t - x = 2pq - p - q - (aq + bp) = (p - 1 - a)q + (q - 1 - b)p \in \langle p, q \rangle. \quad \square$$

Theorem 4 (Hutchings). *If $p \geq 5$ is prime, then after the opening p the second player loses.*

Proof. Let the second player name a legal $q \neq 1$. Then $q > 1$ and $p \nmid q$, so $\gcd(p, q) = 1$. Let $t = pq - p - q = (p - 1)(q - 1) - 1 \geq 3$. By Lemma 3, t is legal after $\{p, q\}$, and every legal move after $T = \{p, q, t\}$ is at most t . By Lemma 1, the player to move from T either loses or wins.

- If the mover loses from T , the first player names t and wins.
- Otherwise there is a legal $x \neq 1$ after T such that the mover loses from $T \cup \{x\}$. The number x is also legal after $\{p, q\}$. By Lemma 3, $t = x + (t - x) \in \langle p, q, x \rangle$, so $\langle p, q, x \rangle = \langle p, q, t, x \rangle$. By Lemma 2, the mover loses from $\{p, q, x\}$, so the first player names x and wins.

In both cases the first player wins after q , so the second player loses from $\{p\}$. $\square$

Corollary 5. *(C1) holds, but (C2) and (C3) fail.*

Proof. The first move 5 is legal, it is not 1, and by Theorem 4 it is winning. So the first player wins, which is (C1). But 5 is prime, hence not composite, so (C2) fails. The prime first move 5 is winning, and after it the second player has no winning response (a position is not both won and lost), so both phrasings of (C3) fail. $\square$

Literature. Wikipedia, *Sylver coinage*, https://en.wikipedia.org/wiki/Sylver_coinage: “Hutchings’s Theorem states that any of the prime numbers 5, 7, 11, 13, ..., wins as a first move”. The strategy-stealing argument with the “ender” t is also described there. The proof above is self-contained.

Readings. The rules are the standard ones: legality is non-membership in the generated submonoid, and naming 1 loses. We formalize (C3) in both of its phrasings, “no prime first move is winning” and “after every prime opening the second player has a winning response”. Both are refuted by $p = 5$, and in fact by every prime $p \geq 5$. Clause (C2) is refuted independently. The conjecture is therefore false under any reading that keeps (C2) or (C3).

4 Lean 4 formalization

The file `lean/Conjecture1213/Basic.lean` (Lean v4.33.1, Mathlib v4.33.1) reuses `Legal`, `Outcome`, `anyNumber` and `not_win_and_lose` from the accepted solution of Conjecture 00000008869 by feiyuceng06-prog (GPL-3.0), with credit. Its main declarations, all in namespace `Submission00000001213`, are:

- `Clause1`, `Clause2`, `Clause3`, `Clause3'` and `Conjecture`: the statement;
- `determined`: Lemma 1;
- `outcome_congr`: Lemma 2;
- `gap_symm`: Lemma 3, via Mathlib's `frobeniusNumber_pair` and `AddSubmonoid.mem_closure_pair`;
- `two_generator_win` and `prime_opening_wins`: Theorem 4;
- `hutchings`: every prime $p \geq 5$ is a winning first move;
- `clause1_true`, `not_clause2`, `not_clause3` and `not_clause3'`.

The main theorem is

```
theorem conjecture_00000001213_false : ¬Conjecture
```

lake build succeeds. `#print axioms for conjecture_00000001213_false, not_clause3, not_clause3', hutchings and clause1_true` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`.